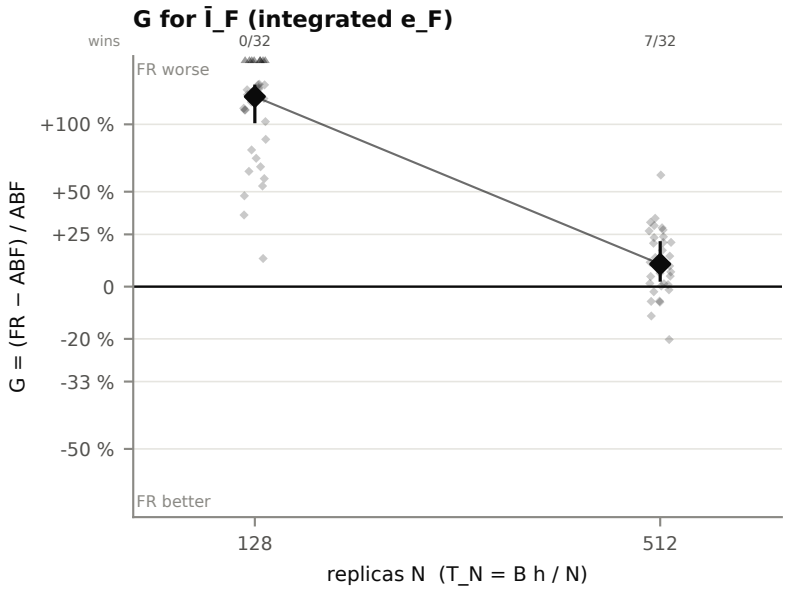
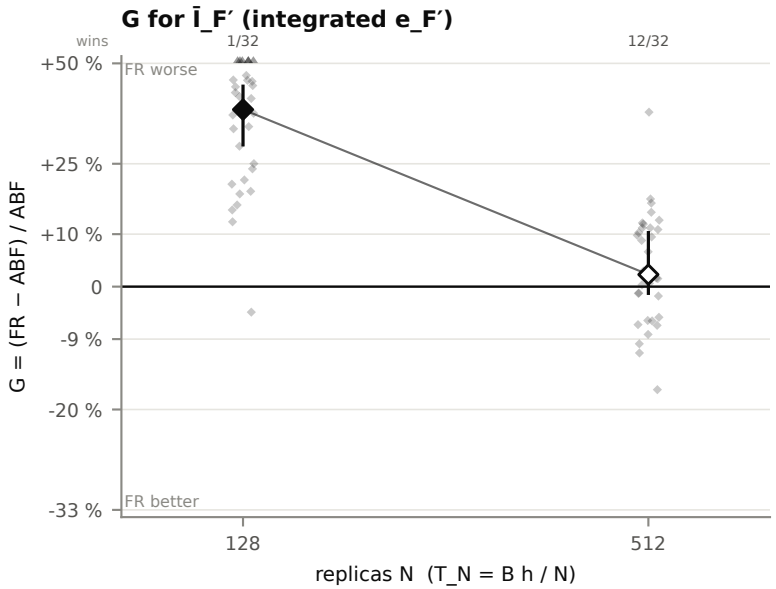
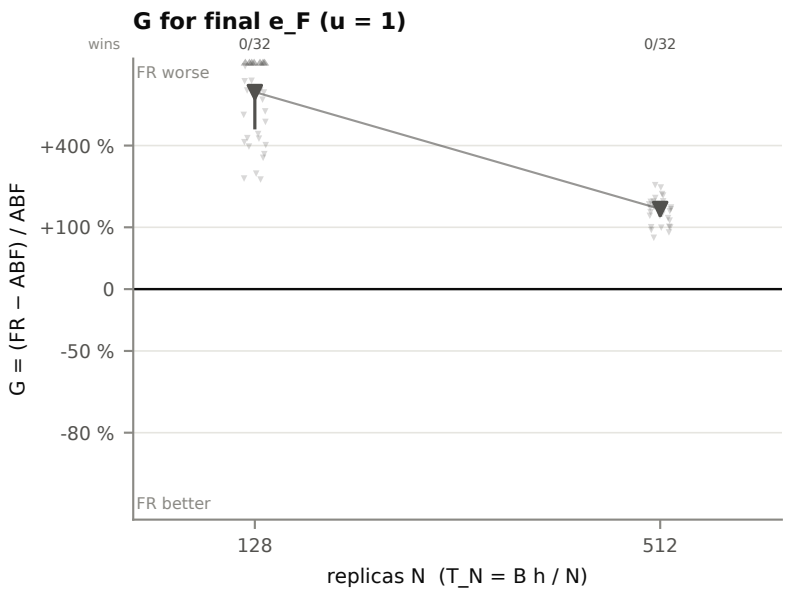
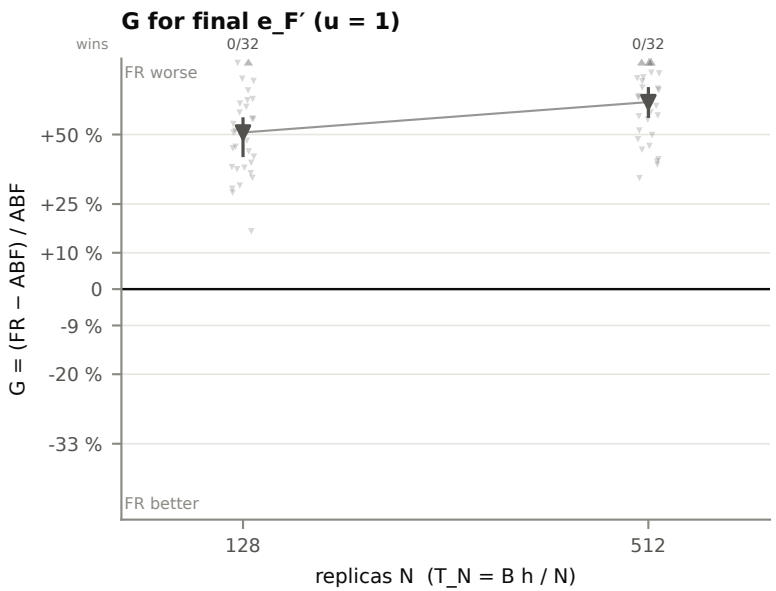
# gateway shift_lam4: FR relative improvement G(N) at equal budget

Equal total budget per arm per seed: B = N × n_steps = 3.2768e+09 walker-steps at every N (h = 2.5e-05, T_N = B h / N); 32 seeds per (N, method).
Paired per seed (same N and seed: shared initial conditions and Langevin noise).




gateway shift_lam4 · S3_fr_gain_vs_N · status: all planned (N, seed, method) complete · FR is not defined at N = 1 (ABF only, by design).
Marker: median of the per-seed G; bar: seed-bootstrap 95 % CI of the median (10000 resamples); filled = CI excludes 0; faint: per-seed G (carets: off scale). Log-ratio axis (FR/ABF) labelled in % change. 'wins' row: seeds with FR < ABF / paired seeds. Red 'FR inactive' / 'no death k/n': FR arms that realised no death (equal to ABF), where G = 0 is FR inactivity, not equivalence.
Source results/final_campaign/expA/shift_lam4/analysis/summary.json (generated 2026-10-11T09:25:19Z, eqb_metrics/2, 10000 bootstrap resamples); plot_synthesis/2.